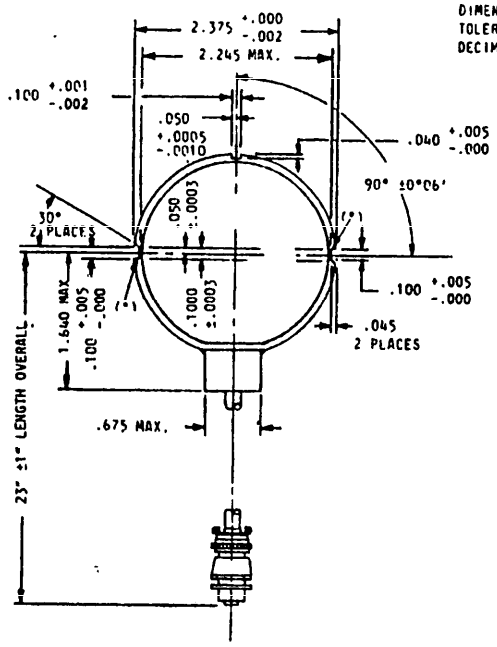
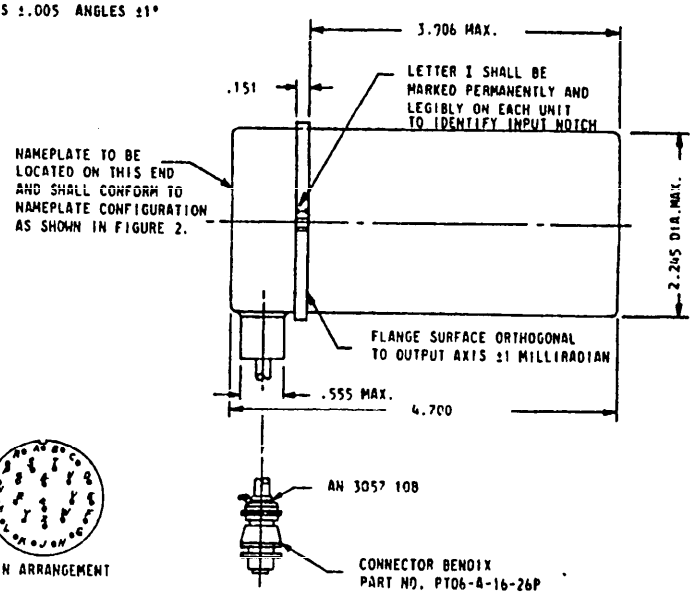
MILITARY SPECIFICATION SHEET
GYROSCOPE, RATE INTEGRATING

THE COMPLETE REQUIREMENTS FOR PROCURING THE GYRO DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-G-81168. IN THE EVENT OF CONFLICT BETWEEN THIS DOCUMENT AND ANY OF THE DOCUMENTS FORMING A PART OF THIS SPECIFICATION, THE REQUIREMENTS OF THIS SPECIFICATION SHALL GOVERN.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ±1/64
DECIMALS ±.005 ANGLES ±1°







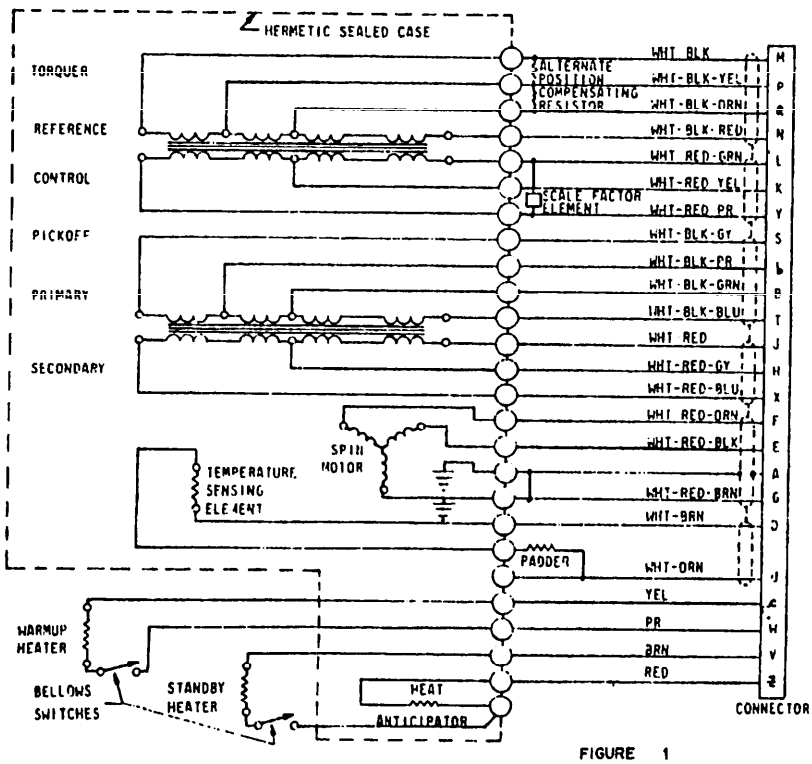
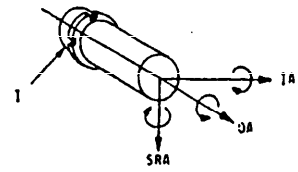


FIGURE 1



NOTES:

1. IT IS RECOMMENDED THAT CONNECTOR PINS M, L T AND K BE GROUNDED FOR MINIMUM CAPACITANCE EFFECTS AT GYRO NULL.
2. CABLE SHIELDS SHALL BE INSULATED FROM THE CONNECTOR HOUD.
3. COLOR CODING OF LEAD WIRES: PRIMARY COLOR - BASE STRIPE - NARROW STRIPE.
4. BELLOWS SWITCHES TO BE OPERATED BY FLOATATION FLUID PRESSURE CHANGE.
5. THE OUTLINE CONFIGURATION IS FOR REFERENCE ONLY.
6. NOTCH ALIGNING SURFACES (1) TO LOCATE INPUT AXIS WITHIN 0.5 MR.
7. THE GYRO SHALL BE WARRANTED FOR 2000 HOURS OF SPINMOTOR OPERATION AT OPERATING TEMPERATURE, OR FOR ONE YEAR, WHICHEVER OCCURS FIRST.
8. THE GYRO SHALL HAVE A MINIMUM SHELF LIFE OF THREE YEARS AT AMBIENT TEMPERATURES WITHIN THE SPECIFIED LIMITS.
9. TEST MOUNTING ACCURACY:
   AXIS ALIGNMENT TESTS ± 0.5 ARC MINUTES
   ALL OTHER TESTS ± 0.5 DEGREES.

FSC-1285 Ⓐ

Ⓐ For changes see pages 1 and 4.

REVISED

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Dielectric Strength: 500 volts rms, 60 cps at the following test points:

J to case	C to case
T to case	V to case
J to T	D to case
L to case	D to C
N to case	D to V
L to N	

Insulation Resistance: Terminal test points same as for Dielectric Strength. Test potential 500 volts d.c.

Spinmotor

Type:	synchronous, hysteresis		
Winding Configuration:	3 phase		
Rated Excitation:			
start	10.5	volts	±1.0
run	9.0	volts	±1.5
	Spinmotor shall run at synchronous speed from 380 to 420 cps.		
Test Excitation:*	10.0	volts	±0.5
	400	cps	±0.1%
	2	% distortion	maximum
Angular Velocity:	12,000	rpm	nominal
Rotor Angular Momentum:	2.0×10^4	gm-cm^2/sec	±5%
Run-up Time:	15	sec	maximum
Power Consumption:	3.0	watts	maximum
	0.5	power factor	±0.1

Pickoff

Type:	four pole, variable reluctance microsyn		
Rated Excitation:	0-100	ma	range
Test Excitation:*	100	ma	±0.5%
	400	cps	±0.1%
	2	% distortion	maximum
Impedance:			
primary	10.3 + j5	ohms	±12%
	with open circuit secondary		
secondary	2170 + j4500	ohms	±12%
	with open circuit primary		
Null Voltage:	10	mv rms	maximum
Phase Shift:	67	degrees	±5
Scale Factor:	26.5	mv/mr	nominal
Linearity Error:	1	%	maximum
	measured at input angles of 0, ±1, ±2, and ±3 degrees.		

*Note: Spinmotor, pickoff, torquer and heater excitations shall be supplied from the same frequency source.

(A) For changes see pages 1 and 4"

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Torquer

Type:	four pole, variable reluctance microsyn		
Reference Winding Excitation:	0-150	ma	range
Control Winding Excitation:	0-150	ma	range
Test Excitation:*	100	ma	±1%
	400	cps	±0.1%
	2	% distortion	maximum
Command Rate Scale Factor:	0.1000	mr/sec/ma^2	±0.5%
	based on ± inputs to 5000 ma^2 inclusive.		

Torquer Linearity Error:

±Input ma^2	Deviation mr/sec	Linearity Error percent
250	.45	.030
750	.85	.057
2000	1.85	.123
5000	4.25	.283
10000	10.25	.683

Maximum Command Rate:	1500	mr/sec	nominal
Impedance:			
reference winding	136 + j280	ohms	±12%
	with open circuit control winding		
control winding	196 + j300	ohms	±12%
	with open circuit reference winding		

Temperature Sensor

Type:	resistance alloy wire		
Scale Factor:	1.65	ohms/°F	±10%
Operating Temperature Resistance:	780	ohms	±1
Induced Voltage:	3.0	mv rms	maximum
Test Excitation:	10	ma	maximum
Operating Temperature	190	°F	±5
Over Temperature Sensor:	Shall be the bellows switch type. Both switches to be located as shown and shall operate at temperatures indicated by the temperature sensing element resistance as follows:		
open	815	ohms	±15
close	785	ohms	minimum

Heaters

Power:			
Warm-up heater	75	watts	minimum
Standby heater	101	watts	±12%
Test Excitation:*	115	volts	±1%
	400	cps	±0.1%

Gyro Unit

Gyro Scale Factor:	35.5	mv/mr	±10%
Input Angular Freedom:	6.0	degrees	minimum
	from pickoff null		

"V" for changes see pages 1 and 4"

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Gyro Phasing:			
Spinmotor:	With an ABC phase sequence, positive spinmotor rotation will result with terminal E connected to line A, terminal G connected to line B, and terminal F connected to line C.		
Pickoff:	Positive output voltage exists when voltage from terminal J to terminal X leads the voltage from T to terminal S by less than 90 degrees.		
Torquer:	Positive torque is obtained when terminals N and L are grounded and the currents to terminals M and Y are in phase.		
Characteristic Time:	3.0	ms	maximum
Axis Alignment:	IA aligned within ±0.5 milliradian maximum of the IRA notch when pickoff is at null. IA shall lie in the plane of the mounting flange mounting surface within ±1 milliradian.		
Acceleration Insensitive Drift: test procedure	60 4.27 of MIL-G-81168	deg/hr	maximum
Acceleration Sensitive Drift: test procedure	45 4.27 of MIL-G-81168	deg/hr	max. resultan
Day to Day Stability:	20	deg/hr	maximum
Ambient Temperature:	-40 to +165	degrees F	range
Operating Life:	2000	hrs	minimum

Vibration, shock, altitude, and moisture resistance shall be conducted in accordance with MIL-G-81168.

Nameplate: Gyros shall be identified with markings conforming to MIL-STD-130. Markings shall be applied by acid, electric etching, permanent marking ink or engraving applied directly to the case. The markings shall be as permanent as the normal life expectancy of the gyro on which it is applied and legibility shall be as required for readability and identification. Markings shall withstand the same environmental tests required of the gyro.

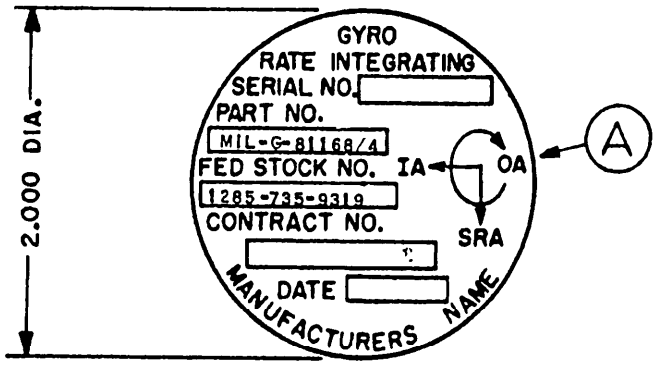


FIGURE 2

NAMEPLATE CONFIGURATION

(A) For changes see pages 1 and 4

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-G-81168/4A(OS) GYROSCOPE, RATE INTEGRATING

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

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